

Adaptive Experience Architecture

The Plain-English Visual Guide to Harness Engineering

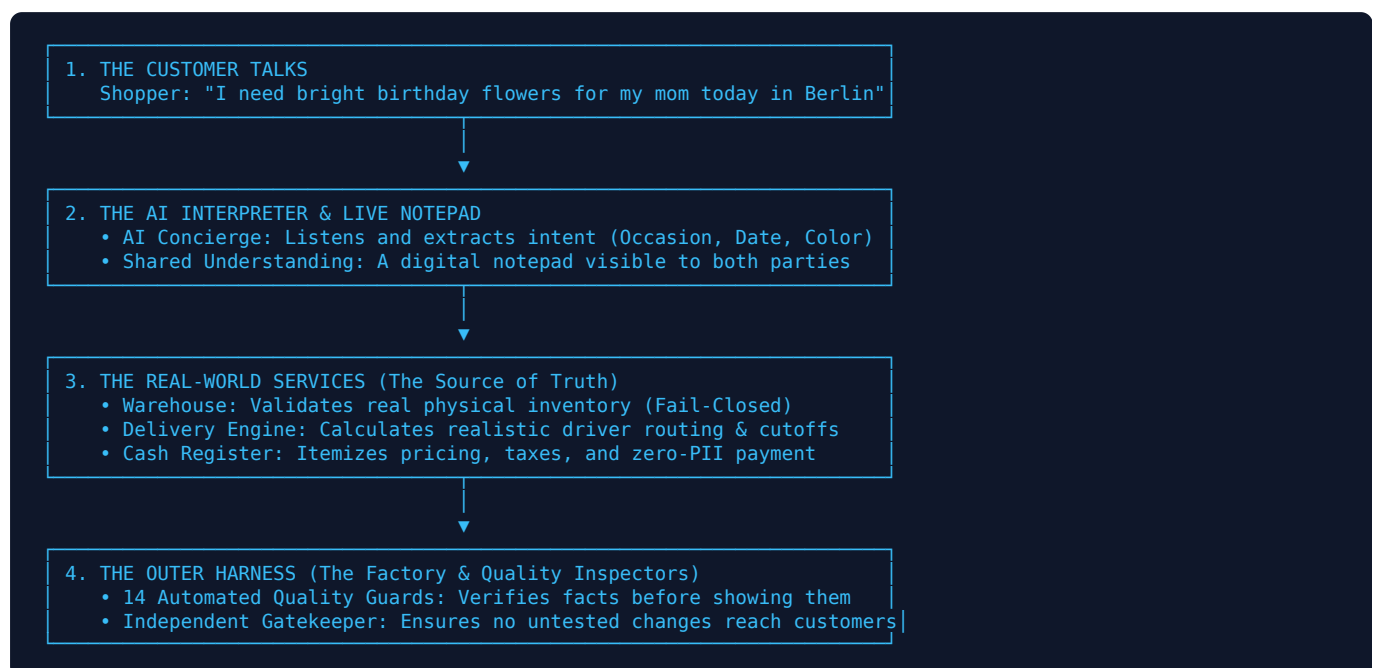
Art of Group • September 2026 • Designed for Product Leaders, Executives & General Audiences • Canonical Reference: aea.artof.link

💡 Why Most AI Prototypes Never Make It to Production

A chatbot that talks charmingly in a demo is not a finished business. When real customers order flowers, the system must check physical cooler inventory, calculate delivery routes, itemize taxes, and process payments without leaking data. AI language models are probabilistic word predictors—they cannot manage a real warehouse. **The Outer Harness** is the software factory built around the AI to guarantee real-world honesty and safety.

1. The Core Formula in Everyday Terms

Adaptive Experience = Shared Understanding + Domain Services + Outer Harness



The Three Golden Rules:

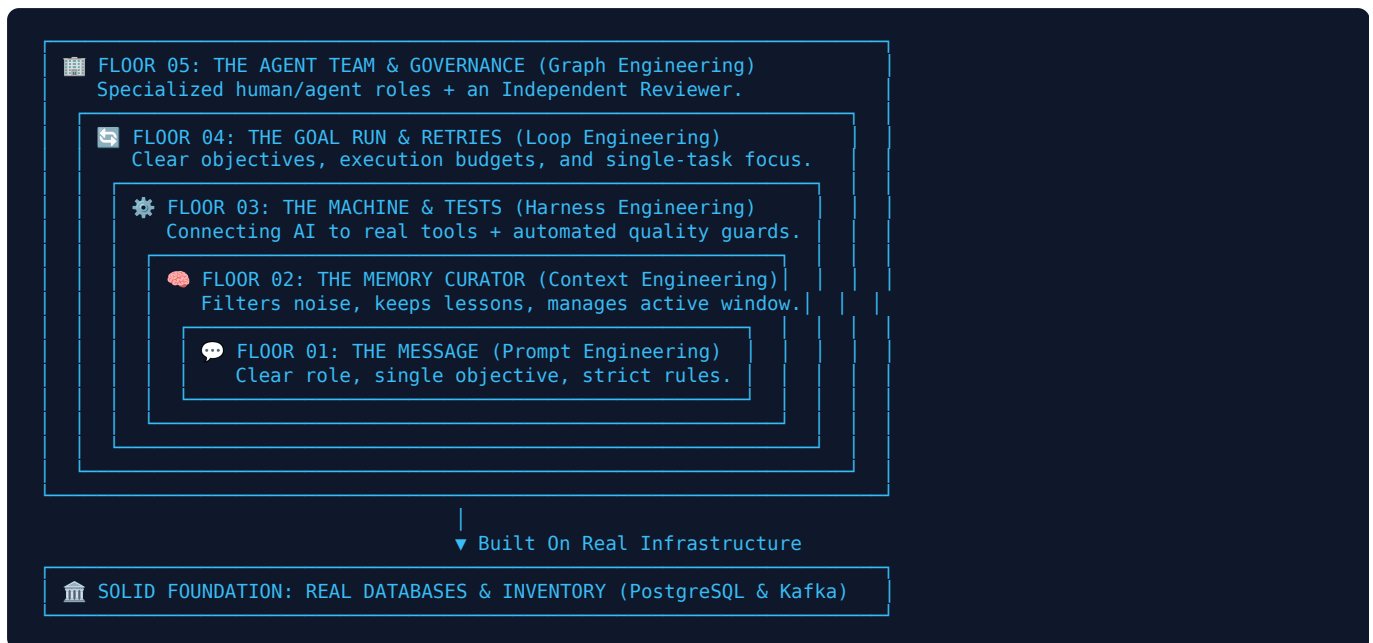
- **AI Interprets, Domain Services Decide:** The AI suggests flowers, but only the database confirms they are in stock.
- **Fail-Closed Availability:** If the inventory server is unreachable, the purchase button turns off. It is far better to say "Checking stock..." than to sell bouquets you cannot deliver.
- **No Self-Approval:** The engineer or AI that writes code is never the one who signs off on pushing it to customers.

2. The Three Eras of Building with AI (2023 → 2026)

Era	Core Focus	What It Optimized	The Fatal Flaw
Era 1: Prompting (2023–24)	Single Utterance	Magic prompt keywords, tone, phrasing	The AI forgets everything the second the chat window closes.
Era 2: Context / RAG (2025)	What the AI sees	Stuffing large PDF manuals and search results	Information overload; the AI knows facts but lacks real actions.
Era 3: Harness Eng. (2026)	The Entire System	Automated test guards, databases, merge gates	None. The AI is bounded by real-world software engineering.

3. The 5 Concentric Floors (Why AI Apps Break)

Think of an AI system like a 5-story building. **Each floor rests on the one below it.** If you skip the lower floors, the top floor collapses.



- **The Dependency Law:** If your multi-agent team keeps failing, don't blame the agents—check your memory filter. Bad input on Floor 2 ruins everything above it.
- **The Economic Law:** Swapping the AI model (like switching from Claude to GPT or Gemini) takes **1 afternoon**. Rebuilding your 5-floor operational harness takes **3 months**. The harness is your real intellectual property.

4. How the "Second Brain" Solves AI Amnesia

Without engineered memory, an AI starts every new conversation from scratch. Dumping hundreds of pages of past chat logs into the prompt makes the AI slow, expensive, and confused. AEA organizes memory into **4 clean vaults**:

1. Procedure Memory (Skills)

Step-by-step playbooks for repeatable workflows (e.g., how to build an Android app or run tests).

2. Correction Memory (Constraints)

Hard rules learned from past mistakes (e.g., "Never invent fake requirement IDs").

3. Relationship Memory (Graph)

Links showing how features, requirements, customer occasions, and code connect.

4. Daily Brief (Handoff)

A clean 1-page summary of exactly where the team left off today so work resumes instantly.

5. Team Organization: 14 Hats Mapped to 6 Functions

Function	Stakeholder Roles	Core Responsibility
1. DISCOVERY	UX Designer, Customer Journey, Support Coordinator	Identify shopper friction, design mobile/web flows, and intake customer needs.
2. STRATEGY	Product Owner, Project Manager	Set business priorities, manage milestone delivery, and enforce scope gates.
3. SAFETY	Security Auditor, Cost Guardian, Performance Guardian, Coherence Guardian	Block prompt injection, enforce cloud spending caps, guarantee fast loading times.
4. BUILDERS	Senior Software Engineer, AI Engineer, DevSecOps Platform	Build backend services, native Android apps, cloud infrastructure, and AI models.
5. GATEKEEPER	Merge Request Coordinator (@aea-mrc)	Independent Review: Verifies all tests pass before code touches production.
6. KNOWLEDGE	Knowledge Guardian (@aea-kg)	Records every breakthrough and lesson into the Second Brain for future sessions.

6. The Six Layers of the Outer Harness in Practice

1. Guides (The Rulebook)

Clear instructions, role limits, and playbooks loaded before starting any task.

2. Sensors (The Smoke Alarms)

Automated tests and fail-closed checks that detect errors before customers see them.

3. The Loop (The Factory Line)

Disciplined workflow: One issue → one branch → one merge request.

4. Memory (The Vault)

Preserves lessons and daily handoffs so past mistakes are never repeated.

5. Permissions (The Keycard)

Strict controls over who can touch sensitive customer data, budgets, or servers.

6. Observability (The Dashboard)

Real-time Grafana telemetry proving the entire system is healthy with hard facts.

7. 10 Sep X-scan (Map Only — Not Shop Evidence)

Three public posts compared to AEA on 10 Sep 2026. x.com was blocked (403); text recovered via fxtwitter. Dual-viewport after CSS stays **Unknown**. This is a taxonomy addendum, not a live Path B claim.

Post	Useful idea	AEA already	We take / we refuse
gipp dual-loop	A second loop writes a small lesson file after the run	Session-end vault + daily brief (optional)	Take: inbox lesson <i>candidate</i> . Refuse: auto-edit skills
beam cost playbook	Audit prompts, cache, match effort; cheaper config vs an eval	Cost Guardian + ADR-016 fail-closed	Take: written playbook. Park: paid hillclimb
avid shared wiki	Provenance; knowledge is not authority; two search paths	Git Second Brain; inbox ≠ canonical docs	Take: keep that split. Refuse: a second Second Brain product

8. Agent Coordination Vocab (14 Sep Keep Learning — Not Shop UI)

Adopted from external learning ([Oxwhrrari X Article](#)) into AEA Outer Harness language via Keep Learning (#441). **Not** a vendor Bot endorsement. No new FR/NFR IDs. Six layers stay the home; this names practices AEA already needed.

Autonomy Ladder (0–4)

Observe → Prepare → Execute-with-approval → Schedule-or-trigger → Coordinate-specialists. Promote by evidence; demote on degradation.

Approval by Reversibility

Finish reversible work. Park send / publish / purchase / delete / production / legal for a human.

Handoff Packet

Pass ownership, not transcripts: from / to / objective / artifacts / decisions / constraints / open_questions / next_gate.

Connection ≠ Authority

Connecting an account is plumbing. A role contract is permission. Keep the two decisions separate.

External name	AEA home	Status
Harness / Loop / Graph	Mapped onto the six Outer Harness layers (schema)	Adopted as vocabulary map
Independent checker	MRC ≠ implementer; probe before status words	Adopted naming (already practiced)
Vendor Bot OS install	—	Refuse as Path B requirement

🌟 **Core Philosophy:** "The engineers who thrive in the AI era are not the ones who write the most code. They are the ones who build the best environments for AI agents and human teams to stay honest."